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INTERVIEW

A Life of Systems Thinking: Michael C. Jackson in Conversation with Matt Lloyd and Rajneesh Chowdhury

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ABSTRACT

This paper is a transcript of the conversation between Michael C. Jackson, Matt Lloyd and Rajneesh Chowdhury, held in 2023 at the University of Hull. The conversation provides a window onto the life and motivations of Jackson, covering his early life and education, his lifelong journey with systems thinking and his hopes for the future of the discipline. Jackson highlights the contribution of systems thinking to those areas where there is greater complexity and uncertainty. He delves into what drove him to craft critical systems thinking as a transdiscipline based on a pragmatist philosophy, understanding the strengths and weaknesses of methodologies from different traditions to aid practitioners in using them in combination. Jackson is optimistic about the future of systems thinking, highlighting that governments and businesses need to embrace it to cope better with the complexities they face.

1 | Background

Matt Lloyd and Rajneesh Chowdhury

Just to start off, a little bit about your background and your work. You've published over 170 papers and chapters, 5 authored books and with one more coming in July 2024. Plus, 7 edited books, 20 PhD students supervised, numerous major projects, and you established the Hull University Business School and led it to Triple Crown accreditation—I (Rajneesh Chowdhury) am very proud of that because I was a student during those days, studying for my MSc. You've also played a leading role in establishing critical systems thinking as a new approach in the field, and you don't seem to stop. What does it feel like to be Mike Jackson?

Mike C. Jackson

You make it sound tiring, and I think it does get tiring as you get older. It's like, you know, when you have a house, and one or two things go wrong with it. You can keep on top of those things. But when 10 things go wrong, you lose track, and you just keep living. It's like that as you get older, and things keep dropping off your body. You just must keep living. Once you get a lot of things wrong with it, you just keep going. And it's not all work. I do, of course, have a lot of hobbies. I'm interested in music. In beer, as you know. And in sports—Hull Kingston Rovers, Hull City, Yorkshire Cricket Club. So quite a lot of hobbies to balance the work life. And I wouldn't be able to do any of it without the support of my wife, Pauline, who looks after the social side of things and family crises.

Matt Lloyd and Rajneesh Chowdhury

Wonderful. What kind of music do you like, Mike?

Mike C. Jackson

Well, let's put it this way, I think I was lucky in being born in an era when music reached a bit of a peak. So obviously the

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Beatles, Rolling Stones, Captain Beefheart, the Grateful Dead, the English folk revival. My particular favourite is Bob Dylan—right through his career. But also, the jazz of the 1950s and 1960s—Miles Davis, John Coltrane and so forth. I won't comment on modern music, just say that I think I was lucky to grow up in that period.

Matt Lloyd and Rajneesh Chowdhury

Wonderful. We've never had a chance to talk about your hobbies. First time. There is a great deal of debate about what systems thinking is and different people interpret it in different ways. So, what is it? Your elevator pitch, for systems thinking: What is systems thinking?

Mike C. Jackson

I like to present systems thinking as a complementary approach to science. In other words, science can deal with many issues—problems when there are a few variables or when there are a lot of variables that you can manage statistically using probability theory. But there are whole areas of life—in fact the most important—where science has little useful to say. Systems thinking makes its contribution in those areas—where there is greater complexity and there are dense interconnections between variables. It complements science in the arena of complex real-world problems involving human beings operating in social systems. That's crucial because it is where decision-makers usually find themselves. Science is good at some things; systems thinking tries to help decision-makers where science is not so good. I'm not sure that I like the notion of an elevator pitch because there are too many different elevator pitches. People try to represent systems thinking in different ways. I think we have to be serious and understand the depth and importance of systems thinking and avoid simplistic elevator pitches.

Matt Lloyd and Rajneesh Chowdhury

We get a bit intrigued when you say systems thinking is complementary to science. So, are we saying that systems thinking is not a science? If you have approaches that help you understand complexity and intervene successfully, surely that is the most sophisticated science, unless you're carrying a different notion of what science is.

Mike C. Jackson

Yeah, I'm talking about the traditional scientific method, about what science can tackle using its traditional method. I don't think I want to describe systems thinking as a science because it has spent all its time trying to deal with the issues that the scientific method can't help you to tackle. If you believe the scientific method is based upon reductionism and repeatable experiments, then very few complex real-world problems yield to such an approach. You can't reduce them to a few causal variables and carry out laboratory experiments. With real-world problems, the circumstances do not stay the same on a second iteration. As soon as you become involved, you change them. So, you can't do repeatable experiments. I do think science and systems thinking inhabit different complementary domains.

Think about COVID-19. Undoubtedly, the solution eventually came through natural science and the invention of vaccines. But the scientific method was unable to do anything in terms of predicting the way that the epidemic would develop because too much information was missing, in the early days, to feed into the epidemiological models and those models could not predict what politicians would do and how the populace would react. Government wanted to 'follow the science'. But the science was not up to the task. I think that if decision-makers had the variety of systems approaches at their disposal, they would have prepared better and responded better to what happened during the COVID-19 pandemic (see Jackson 2020).

Matt Lloyd and Rajneesh Chowdhury

Sure, thank you. Next, we always wondered how you got into systems thinking and how were you introduced to it for the first time.

Mike C. Jackson

I always look back at being taught geography at school—'A' level geography. They used to teach it by dividing countries up into regions and describing what happened in each of those regions. And I remember at the time thinking there's something wrong with this—you can't really understand the regions independently. You must look at them in interconnection with other regions, with the country, with the world. So, I had doubts about what I saw as geographical reductionism at that time.

When I went to university, I did politics, philosophy and economics (PPE) and I moved towards the sociological options and came across formal systems thinking there for the first time, in the work of people such as Parsons and Marx. It being the 1960s and early 1970s, and me being a radical, my interest in sociological systems thinking continued while I was working in the civil service after university. I probably spent more time reading systems classics than I did doing my job in the civil service. Eventually this interest came to dominate, and I went back to university, choosing the 'Systems in Management' course that was being run at Lancaster University. Peter Checkland was the key figure and, of course, I came face to face with soft systems methodology.

Checkland's soft systems methodology impressed me as a sophisticated attempt to translate systems ideas into practice. At Lancaster, you tended to learn just about soft systems thinking; I spent a year learning about that approach. It was only later that I went on to study other systems approaches. But I certainly got an excellent background in systems at Lancaster. Peter Checkland was a brilliant teacher and mentor. I intended to go back into the workplace after Lancaster, but I became so interested in the systems approach, and applying systems thinking, that I started the move to becoming an academic, beginning a PhD at Warwick.

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Matt Lloyd and Rajneesh Chowdhury

We're interested about your time at Lancaster, and having read quite a lot of your work, there seems to be quite a lot of reference to Burrell and Morgan who were active at Lancaster at the time. It seemed to have been quite a significant influence on you. Were they part of what was going on with the development of soft systems methodology, with Peter, or just an excellent thing going on at the university at the time that helped?

Mike C. Jackson

It helped that there were links between the Department of Behaviour in Organisations, as it was called, and the Department of Systems. Burrell and Morgan were active in the Department of Behaviour in Organisations. I vaguely knew Gareth Morgan. It was a great time to be at Lancaster. I was there in 1977–1978. Burrell and Morgan's (1979) 'Sociological Paradigms and Organisational Analysis' came out in 1979 and Peter Checkland's (1981) 'Systems Thinking, Systems Practice' in 1981. Checkland's book has a section where he relates his soft systems methodology to the Burrell and Morgan paradigm grid. You can see that important interchanges were going on. You had social theory informing the development of systems thinking at Lancaster at that time. That's the sort of experience I got, and having a background as a social scientist, I embraced that tradition and have tried to extend it.

Matt Lloyd and Rajneesh Chowdhury

OK, the final question in the first section on your background. The job of a successful academic combines three big independent jobs in itself—researcher, lecturer and manager. And I (Rajneesh Chowdhury) saw from my own experience, when I was at Hull, that you successfully combined and excelled in all three. How did you manage—what's the secret sauce to do that?

Mike C. Jackson

Well, you're very kind. I sometimes wonder whether I'd have been better off concentrating on one rather than doing all three. I wonder how I would have done if I'd just been a university administrator. How I would have done if I had just been a researcher. Clearly, I would have had more time to concentrate my energies if I'd just focussed on one. But I don't regret it now because I think what I learned about systems thinking, and using it in consultancy, helped me as a Dean. I've written a paper on how I used critical systems thinking throughout the period when I was a Dean (see Jackson 2019, 621–632). And, I wouldn't have felt as confident as I do proclaiming the usefulness of systems ideas, and how you can employ them, if I hadn't practised them as Dean over a relatively long period of time. Over the 12 years I was Dean, Hull University Business School became quite a substantial enterprise. When I left, in 2011, it had around 180 staff and a turnover of about £28 million. So, I was managing a medium-sized enterprise and using systems ideas in that context over a long period. I continued learning how useful they were in helping with practical day-to-day management. I think being a Dean and a systems thinker complemented one another. My experience as Dean gave me more confidence in saying what I thought was good about systems thinking and the systems thinking helped me to be what, I hope, was a successful Dean. Turning to teaching, I think it undoubtedly benefited me having had a spell outside academia between university and coming back into university life. I think if you go straight through the university system, you tend to pick up some rather jargonish language and can get carried away with that language, expecting other people to translate what you're saying into what they can understand. I hope that I've never been like that. I've always felt that I had to make my ideas intelligible to the people that I was speaking to. And I have tried to do that with undergraduate, postgraduate and postexperience students.

2 | Systems Thinking and Critical Systems Thinking

Matt Lloyd and Rajneesh Chowdhury

OK Mike, we would be interested now to learn a bit about your interest in the subject matter of systems itself, always very varied, since you first came into the field. Who were some of the key scholars who've inspired you and how do you see the subject evolving?

Mike C. Jackson

How my own thinking evolved? Well, as I say in the early days it was an interest in sociological systems theory, based upon Parsons and then on Marx. And then I became involved, after Lancaster, in the application of systems ideas and so that became important. There was a shift there in what interested me. Because I thought systems thinking was a really good way of intervening and improving the world that became important. At Lancaster, I was taught soft systems methodology. I must admit I don't think I particularly understood it until I got out into the world and tried to use it, both with student projects and in consultancy. And then I became to fully grasp the sophistication and the strengths behind soft systems methodology and the work that Checkland and his colleagues had put in to develop it.

But I also made it my business to have a look around at other systems work. I think von Bertalanffy's (1971) collection of essays 'General System Theory' remains a remarkable book. I wish more people would really read it rather than just commenting on it, or just referencing it. It's about emergence as much as it is about the concept of a general systems theory. Its total opposition to reductionism is central to its message—an opposition to reducing biological phenomena to mechanism, human phenomena to mechanism or to biology and social phenomena to a lower level of complexity than they exhibit.

I became interested in Stafford Beer's work. I still think 'Brain of the Firm' (Beer 1981) is a superb book from which we can learn so much. It brings all the ideas of cybernetics together

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in a practical, useful tool that people can use to improve organisations, ensuring their viability and effectiveness. It's just incredible the job he did, drawing, of course, on an analogy between an organisation and the human brain and nervous system. And I was fascinated by the work he did in Chile for the Allende government.

Russ Ackoff was a very significant mentor. I got to know Russ well. I've often told the story of when I met him at a conference in Washington. I had just written the article criticising the work of Churchman, Ackoff and Checkland (Jackson 1982), and I said to Russ: I've just written this article, have you read it? What do you think of it? And he said: I don't think you've understood anything I've ever written. He could be quite brusque. But we became good friends, and he was very supportive. The pithiness of his writing and his ability to get difficult ideas over concisely and in a way that managers and leaders can understand was a great skill. I think he was right on almost everything he said about systems and systems thinking. Like Churchman, he was an early critical systems thinker. He saw the strength of mechanistic and biological analogies in their place but thought that society had moved beyond their ability to understand what was happening and that we now need to base systems thinking on social systemic thinking. His work was very important.

I tried to understand the value of all the different systems approaches. Systems engineering has great strengths, of course, and was central to the success of the Apollo project. It has much to offer as a way of integrating components and subsystems into a whole to achieve a goal. System dynamics and critical systems heuristics have their strengths as well.

So, then my thinking moved on. It seemed that systems thinking had splintered. We have a variety of different systems approaches. I began to wonder how we could recreate the power and the unity that systems thinking seemed to have in the early days. How could we make it coherent again as a transdiscipline? So that's when I started to develop the critical systems thinking research programme.

Matt Lloyd and Rajneesh Chowdhury

Thank you for introducing the concept of critical systems thinking. In the last few years, you've increasingly been talking about critical systems practice. And you say that critical systems practice is pragmatising critical systems thinking. Can you tell us a bit more about critical systems thinking, critical systems practice and pragmatism.

Mike C. Jackson

OK, critical systems thinking: Let's start with the period in which it was born. In the early 1980s, there was this variety of different systems approaches. And it's not just that there was a variety, it's that the advocates of different approaches had become different factions at war with one another. There was a paradigm crisis; a Kuhnian paradigm crisis. It looked as though the old status quo based upon hard systems approaches, such as systems engineering and systems analysis, had broken down. But there was no alternative paradigm that had taken its place. And so, the transdiscipline of systems thinking was in a period of crisis. The job of critical systems thinking, I thought, was to put things together again. It sought to do this by carrying out a critique of all the existing systems approaches—a job it carries on to this day. This critique involved determining what each was good at and not good at—what their strengths and weaknesses are. Once you understand what their different strengths and weaknesses are, you can contemplate using them in combination. The strengths of certain approaches can be used to overcome the weaknesses of others. The best approach to improving issues and solving problems turns out to be using the different approaches in a pluralistic way, in combination. Critical systems thinking was also interested in ensuring that issues of disadvantage and discrimination were on the agenda and was dedicated to ensuring fairness for groups that had no representation in decision-making or were prejudiced against in society. It was inevitable that we should ask: How do you translate the ideas of critical systems thinking into critical systems practice?

Critical systems practice really got going when Bob Flood and I came up with the notion of 'total systems intervention' in the 1991 book 'Creative Problem Solving: Total Systems Intervention' (Flood and Jackson 1991a). Incidentally, that's a title that we both now think is the worst possible that we could have come up with. It is preposterous to think you could do a 'total systems intervention'. I think critical systems thinking is all about understanding you can't do that, as indeed the content of the book is clear about. Anyway, a solid start was made in thinking through how you translate critical systems thinking into practice. In what I now call critical systems practice, it's clear that you need to find a way of honouring the commitments of critical systems thinking—to critique, pluralism and improvement—at all stages of an intervention. You don't want to look at the world through one metaphor or systemic lens. They can all be critiqued for being limited. You must view the world through a variety of perspectives because there's no universal truth. Then, you must bring to bear the different methodologies that correspond to those different worldviews, because you are never absolutely sure which of them is going to gain traction in the situation that you're confronting. And you should be willing to shift approaches during a project as priorities change and as your own intervention changes the situation with which you're engaged. There must be a pluralism of methodologies at play. And when you're evaluating how well you've done in that intervention, what improvements you've made, again you must evaluate from a pluralistic perspective. You must check whether you have improved efficiency and efficacy; whether the system is more agile and viable over time; do the people in the situation understand each other better, have enhanced mutual understanding; check on whether the systems intervention has benefited those who are affected even if they were not involved in the decision-making; check on potential unintended consequences. You need a pluralistic, multidimensional kind of evaluation. That's an example of remaining true to the pluralist commitment of critical systems thinking in critical systems practice. Critical systems practice seeks to honour those ideas behind critical systems thinking and translate them into practice.

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Matt Lloyd and Rajneesh Chowdhury

So, is that what is called pragmatism?

Mike C. Jackson

I think that critical systems thinking was very slow in identifying a philosophy that it was comfortable with. And I always thought that was a bit of a weakness. So, in the early days of critical systems thinking the idea was, based upon the work of Habermas, that you could stand back and allocate the different systems methodologies to different roles. The harder approaches could serve a technical interest in increasing prediction and control. The softer approaches, like soft systems methodology, could serve a practical interest in improving mutual understanding about what you were trying to achieve. Work such as Ulrich's (1983) critical systems heuristics, the more radical approaches, could serve the emancipatory interest, representing the interests of disadvantaged groups. I no longer think that this kind of metaparadigm approach is defensible because there is no God's eye view. You're locked into the situation yourself. Each of the paradigms, on which the methodologies are based, claims to explain everything. They give you different explanations of the situation. It is difficult to choose between them. Once you get to that point, you start to wonder whether we can ever find philosophical support for critical systems thinking. Some people have tried to say that it must be a paradigm itself, and we just have to discover a paradigm capable of housing its various commitments, especially pluralism. Gerald Midgley (2000) suggested a process paradigm; John Mingers (2014) argued that critical realism is well suited to supporting critical systems thinking. The problem is that many people in the systems tradition don't agree that their ideas fit well into either process philosophy or critical realism. I preferred to accept the existence of different paradigms as central to critical systems thinking and to argue that the way forward was not to privilege one but to use them to confront one another. So, if you know you're operating within the functionalist paradigm, using a hard systems approach to improve prediction and control, you need constantly to swap into another way of thinking to see what you were doing from another perspective. One approach allows you to see certain things clearly, and do certain things, but there'll be blind spots. So, you must always be willing to swap to another way of looking at the world using another systemic lens and potentially employ another systems methodology. If you are only concerned with efficiency and efficacy, you might eliminate redundancy and, therefore, threaten resilience. You might damage relations between stakeholders. You may disadvantage certain groups by the actions that you're taking. You can only identify these things clearly by adopting a different standpoint. Therefore, I advocated the use of different paradigms, allowing people to inhabit these different perspectives. They had to be used in combination and to confront and challenge each other.

But I still wasn't happy because that seemed to lead to a relativist position. These paradigms all have something to say. How do I know which is going to be useful for a particular set of circumstances? And how am I going to learn which are most useful for what over time? Thanks to the work of Zhichang Zhu, a colleague at the Centre for Systems Studies, that I think I have found a way forward. This involves embracing philosophical pragmatism. I really had, in my mind, a revelatory moment. I think a lot of people, once they first start reading the pragmatist classics, get the same sort of feeling.

Pragmatism offers a way of understanding things which is totally different to traditional philosophy and to an extent to traditional science, although new conceptions of the scientific method are catching up. Because, with pragmatism you can accept that a variety of perspectives are going to be potentially useful to you without fully embracing any of them or falling into relativism. You're not asking the question anymore: Does this theory correspond to some reality out there? Does it yield eternal truths? You're not trying to measure your thinking in terms of how closely it corresponds to reality—which I think is an impossible task. Instead, you're asking: If I see and do things this way, does it bring about some sort of useful improvement for the human species in the here and now? Truth is no longer a function of whether this paradigm is right or wrong, whether it provides us with an exact picture of the world. Pragmatism says abandon the 'spectator theory of truth'—the idea that truth is about building a model which captures the nature of reality. How could we possibly do that? We've evolved as a species. We are embodied minds. We're brought up in particular cultural traditions, speaking certain languages, from which we cannot readily detach ourselves. So, truth does not come from mirroring nature but is what we can agree on as a community that will lead to improvement for the human species at this moment in time. And, of course, I would add that improvement for the human species also involves improvement for other species and for the environment that we inhabit as a human species.

For me, therefore, the latest and most significant change in critical systems thinking and critical systems practice is certainly the embrace of pragmatist philosophy. We must abandon the spectator theory of truth. Truth is historical. It is different at different periods and is conditioned by our ability as human beings to understand things. Truth is about what's useful for us. And we can make truth happen. In the social domain, we can design situations which correspond to the truths that we hold dear. I think this is liberating and I think it corresponds to what critical systems thinking has always been about. But it has never perhaps, until now, fully grasped what it's about. It can do this when it aligns itself to pragmatism.

Matt Lloyd and Rajneesh Chowdhury

Wonderful. That's very enlightening, Mike. Thank you. You know, when you were speaking, we were also wondering, isn't it asking too much from a practitioner to do critical systems practice, to be aware of so many methodologies, paradigms and multiple success factors. Aren't we asking too much from a practitioner?

Mike C. Jackson

Well, the way things are, maybe we should be asking a lot. Traditional, reductionist, mechanistic ways of thinking don't

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seem to be taking us very far. You only need to look at all the different things which are going wrong in the world, and I won't enumerate them because there are too many and they are too frightening. You cannot say that the way we think now, the way we do things, contribute to benefiting people in the way we have the capacity to benefit them given the scientific and technological advances we have made. We need to think and act differently. So why not hold up a strong challenge to decision-makers and practitioners? Challenge them with alternative ways of thinking and insist that if they genuinely want to improve things, then they will have to take on and learn ways of thinking differently about the world and the way they want to change it. That said, critical systems thinkers clearly must meet people who have experience, knowledge of a situation, well-grounded common sense, halfway. I would never say systems thinking can replace experience and common sense. I don't pretend critical systems thinking has all the answers. I do think it can help people with the knowledge of the situation to improve the way they think about it and act in it. It's an aid to decision-making and common sense. In what ways can it help?

The work I've been doing on critical systems practice has sought to look at those ways of thinking about the world and those ways of acting in the world that have proven useful to the human species over time. Critical systems practice is based upon a notion that over time, and this comes from Stephen Pepper's work, there have been certain 'world hypotheses' which have genuinely assisted the human species in making its way in the world, both in taming the physical world and in building viable sociocultural systems. In addition to Pepper's (1942) world hypotheses, I considered what sociological paradigms and 'metaphors of organisation' (drawing on Gareth Morgan 1986) have stood the test of time in proving themselves to be helpful ways of looking at the world and engaging with it. There are lots of overlaps and you end up with a relatively small number—five to be precise—of 'adequate' systemic perspectives that give you a good handle on how things work. It's always useful to think about the world using a mechanistic lens. Where would we be as a species if we didn't have this systemic perspective to help us enhance our ability to predict and control things? It yields powerful insights to employ an organismic systemic perspective, asking questions such as: How do things fit together? How should the parts function to maintain a whole in existence? How can a system be designed to adapt in a turbulent environment? It's worth thinking about the world in a contextualist way: How are people thinking about what is happening? How can we create mutual understanding? Are there alternative ways of thinking about it? How can we challenge existing ways of thinking? It's powerful to bring a societal/environmental perspective to bear: Who is being disadvantaged? Can we not do better for everyone? Are we giving due consideration to the environment on which we all depend? And, finally, it's also worth thinking about the world in terms of interrelationships. How does one thing affect another and if you do this now how it might have long-term consequences somewhere else?

That's it. Five ways of thinking about the world highlighted as a basis for critical systems practice. Not, I think, too complex or difficult to get your head around. And, we happen to have developed five types of systems approach that can respond to those different ways of viewing the world—hard/technical methodologies, living systems, soft systems, emancipatory systems and system dynamics methodologies. So, if you're looking at an organisation through a mechanistic lens and it's working quite efficiently, but it is clear, from an organismic perspective, that it's not adapting well to its environment, you can do something about it. You can use something like Stafford Beer's viable system model to address the concerns raised by the organismic perspective. You must keep all the systemic lenses and methodologies in play, drawing attention to different things and being able to guide action in different ways as appropriate. And it's not a chance that there are five world hypotheses which have stood the test of time in helping human species to cope with the world, and that we happen to have five types of systems methodology, well researched and tested, corresponding to those world hypotheses, and supporting them. Rather, it's a correspondence that points to the link between our abilities to do well in the world and these five ways of thinking and acting.

Matt Lloyd and Rajneesh Chowdhury

Last question in this second section of this interview. You seem to be writing a lot and talking a lot about Bogdanov's work. We remember when we were students, we were never exposed to Bogdanov's work but he seems to be a thinker that's emerged as a significant figure in systems thinking. So why do you think Bogdanov's work is so important?

Mike C. Jackson

I think it's important to acknowledge that Bogdanov was the first rigorous systems thinker and a critical systems thinker. Of course, you can quote certain pre-Socratic philosophers, or Aristotle on the whole being more than the sum of its parts; you can see Hegel and Marx as systems thinkers. But Bogdanov was the first rigorous systems thinker, anticipating many of the ideas later found in the writings of von Bertalanffy and Wiener. He is best known for his 'tektology', the universal science of organisation. The links with von Bertalanffy's general system theory and Wiener's cybernetics, the science of communication and control in all types of systems, are obvious. But I think that's the least useful part of his work. The idea that you can find universal laws of systems that are applicable whether the system is mechanical, biological, human or social, throws up some nice analogies and metaphors but I don't think, in the end, gets you very far. If we believe we'll ever get general system laws, a universal science of organisation, then I think we're kidding ourselves. We won't. It's like waiting for Godot. It's not going to come. I'm sorry that a lot of people seem to be wasting their time on that.

I think the biggest contribution systems thinking can make is to practice. What is really important about Bogdanov is the kind of pragmatist philosophy that he developed before his 'science' of tektology. Both Bogdanov and William James were influenced by the Austrian philosopher Ernst Mach. Mach was a physicist- philosopher who also, incidentally, influenced Einstein and Heisenberg. Bogdanov's philosophy is very much a pragmatic approach from a systemic perspective. So, in developing critical

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systems thinking along pragmatic lines, I necessarily look back to that very early pragmatic approach within systems thinking that Bogdanov was championing.

Bogdanov was at one time a leading figure in the Bolshevik party in Russia prior to the revolution—equal in stature for the time with Lenin. They fell out because Lenin believed that the truths of Marxism were objective. They were there for ever. Marx's and Engels' work was seen as constituting a science which could describe the way societies inevitably would develop. There was objective truth for Lenin. And Bogdanov disagreed. He thought that truth was historical, and we need to continually enhance our thinking considering, for example, the discoveries of modern science. For Bogdanov, truth is created in a particular historical era, reflecting particularly historical conditions. And for Bogdanov, as for the pragmatists, truth is something that proves useful to the human species. He was interested in finding historical truths that would enable people, the proletariat in his time, to become successful world-builders, capable of creating the kind of society that they wanted to see come about. I think that's interesting. And because I think that pragmatism is the route forward for critical systems thinking, it's great that there was a person of Bogdanov's standing who was the earliest of the rigorous system thinkers and in his philosophical work, at least, a pragmatist.

It's true that his earlier philosophical pragmatism doesn't sit easily with his later tektology. But, then again, von Bertalanffy's perspectivist philosophy doesn't chime well with the search for a mathematical general system theory. von Bertalanffy insisted that he was a perspectivist philosopher, valuing different perspectives, different worldviews. He also comes close to pragmatism. And it's the pragmatist stuff in Bogdanov and von Bertalanffy which I think continues to be most useful—an inspiration for those who want to develop systems thinking in a practical way and who see its biggest achievement as being the development of a range of systems methodologies capable of addressing complexity in different ways.

Matt Lloyd and Rajneesh Chowdhury

Just to clarify, did you say von Bertalanffy's work can be seen as pragmatist?

Mike C. Jackson

I think if you read 'General System Theory' carefully, you'll find a lot in von Bertalanffy which is pragmatist in orientation. He was much influenced by Gestalt psychology, for example. Once you start becoming interested in Gestalt psychology, you realise that the mind has an important role to play in what you see in the world. von Bertalanffy talks about biology having an influence upon the way we see the world, psychology having an influence upon the way that we see the world, language having an influence, culture having an influence. You can't expect people to think about things in the same way. And so he believes that general system theory is just one powerful way of thinking about the world. And that the main benefit it has is that it can help improve the human condition.

3 | Criticisms of Critical Systems Thinking and Practice

Matt Lloyd and Rajneesh Chowdhury

All right Mike. Let's talk about some of the critiques of your work. You have crafted *EPIC* (Explore, Produce, Intervene, Check) as an overarching framework for critical systems practice. One of the criticisms of *EPIC* is that it comes across as a linear methodology that is unfit for complex real-world challenges. How do you react to that?

Mike C. Jackson

I think people are misusing the concept if they say that *EPIC* is linear. I don't understand that criticism at all. It's true that, to explain how you might go about using the approach, it's divided into four stages. But the fourth stage, where you're evaluating how well an intervention has gone through different lenses, so you're getting a rounded appreciation of what's been achieved, is very similar to the first. Reviewing things in terms of efficiency, sustainability, mutual understanding and so forth is almost the same as the first stage—when you're considering a situation for the first time and asking: If I look through these different lenses, what might be the key issues for me to seek to improve? You are clearly going round a cycle of intervention if the final stage is almost the same as the first stage. And beyond that, the commitments of critical systems thinking are honoured at every stage of the approach. Your mind is being kept open to new possibilities at every stage. You're constantly critiquing what you're doing in terms of other ways of looking at things or other ways of doing things. There's an ongoing debate, if you like, horizontally at each stage, as well as in the circular use of the approach. I don't understand how it could ever be labelled as linear.

Matt Lloyd and Rajneesh Chowdhury

And from our own reading of your work, you have always depicted it as a circle, and you've used the word 'iterative' many times. We've never seen it as a linear approach.

Mike C. Jackson

I don't know how many times you need to use the word 'iterative' to get that message across, but people have a penchant for criticising things as linear approaches. The fact is we must try and represent, somehow, what it is we're getting at and, if you do it in prose, then obviously it can be condemned as linear. If you do it in a diagram, you try and put it in a circle and still people seem to say that it's linear, so it's a bit hard to get around that criticism if they're insistent upon it, but I can't see it myself.

Matt Lloyd and Rajneesh Chowdhury

You have also been recently criticised for claiming that critical systems practice offers 'the best' approach for evaluation. How

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would you react to that? And we kind of agree with that criticism because when you talk about critical systems thinking and practice, it's about being critical and surely that should extend to your own work.

Mike C. Jackson

Well, I think the clue is in the mnemonic *EPIC*. I don't think I would ever claim anything as 'epic', apart from maybe if Hull Kingston Rovers won the Rugby League grand final. It's meant to indicate that what we're talking about here is an ideal type, in Weber's sense. In other words, everything being equal, everything going your way, no constraints, no opposition, if you wanted to do a critical systems practice evaluation, then based on the knowledge we have currently about systems thinking this is how we would do it. It's an ideal type that you would never be able to replicate exactly in the real world. It's like Weber's ideal type of bureaucracy. You would never actually find any organisation that mirrors exactly all those features. He just tried to give you some sort of useful model that can help with analysis of what actually exists in the 'real world'. *EPIC* is an ideal-type model of how, if the circumstances were perfect, you would do a critical system practice evaluation—making sure you are considering all the different systems approaches and all the different worldviews that they encapsulate. Now, clearly, that's not going to be feasible in the great majority of cases because of constraints on what's possible. In a particular situation, you have to adjust the ideal type to get the best outcome you can.

Michael Quinn Patton, in the debate I had with him about evaluation (Patton 2023; Jackson 2023), seems to have interpreted the ideal type as the best in all circumstances. That's not what I was saying. He wants to come from the opposite direction. He assumes the constraints on a full critical systems practice evaluation are insurmountable and asks only that we try to introduce a few systemic concepts to make the evaluation more systemic. He thinks that's the best we are likely to be able to do. To me, that's not good enough. People will interpret those concepts according to their favoured way of looking at things—probably reverting to the dominant mechanistic mode. They require a bit more guidance than that to carry out a proper systemic evaluation which, for me, must be multiperspectival and multimethodological. I think the guidance provided by an ideal type is exactly what they need, to enable them to reflect on where they are falling, perhaps necessarily, short in any practical application. I don't think *EPIC*, as an ideal type is a negation of being critical. It provides a means to be critical because it enables you to see where you might be falling short in carrying out what may well still be the best evaluation achievable in a particular context.

4 | The Future of Systems Thinking

Matt Lloyd and Rajneesh Chowdhury

We'd now like to turn to the future of systems thinking. The first thing that comes to mind is that you, along with Bob Flood, first introduced the concept of 'waves' of systems thinking. And ever since, there has been a lot of debate about whether the wave metaphor is useful or not. And nowadays we hear more and more about a fourth wave as well. Just wanted to have your opinion about what you think of the 'waves' idea in general. And the credibility of a fourth 'wave' that, in some circles, systems thinkers are talking about.

Mike C. Jackson

Yeah, Bob and I came up with it in 'Critical Systems Thinking: Directed Readings' (Flood and Jackson 1991b). I now think it's an unfortunate analogy because it's very misleading. And I'm sorry that some people still propagate it. There are different strands of systems thinking, true, but I don't think they developed as waves in the way that is talked about now. For this reason, people talk about the first wave as consisting of hard systems approaches within which they include systems engineering, system dynamics and Beer's viable system model. But those systems approaches are so very different in the assumptions they make. For example, Beer's work draws far more on biological analogies than do systems engineering and system dynamics. They're so different that it's misleading to see them as constituting a 'wave'. They're different in the assumptions they make, the concepts they employ, and they are useful for different purposes.

Secondly, the 'waves' metaphor tends to suggest that one 'wave' comes in and then retreats and then another 'wave' comes and replaces it. I think elements of all three so-called 'waves' were there at the beginning. You've got Boulding (1961) emphasising 'the image'; Churchman (1979) stating that the systems approach begins 'when first you see the world through the eyes of another'. Versions of soft systems thinking, apparently the essence of a second 'wave', were there from the beginning. Think, as well, of the pragmatism of Bogdanov and the perspective philosophy of von Bertalanffy. It's not as though a second 'wave' has replaced the first 'wave' and a third, critical 'wave' has replaced the second 'wave'. And, of course, they are all still there. Indeed, the largest communities in the systems field are still systems engineering and system dynamics, which people seem to be saying washed over us as part of the first 'wave'. The notion that there was a 'wave' of hard systems approaches, replaced by soft systems approaches, replaced by critical systems approaches is, I think, just a nonsense. Finally on this matter, in the so-called 'third' wave, emancipatory systems approaches, dealing specifically with marginalisation and disadvantage, are lumped together with critical systems thinking. That too is simply wrong, because critical systems thinking is an attempt to appreciate the strengths and weaknesses of what all the 'waves', if we must, have washed up. I don't like the 'wave' metaphor anymore. It is lazy thinking, and I would encourage people to call it out whenever they see it being used.

As regards a fourth 'wave', there doesn't seem to be any clarity about what that is. I've seen different versions. One suggestion is that it considers not just rational thinking, in the Western tradition, but other ways of addressing issues. It encourages us to embrace Eastern and Indigenous ways of relating to the world. Another suggestion is that the fourth 'wave' attempts to simplify systems thinking. How can simplifying things constitute an important new direction for systems thinking? It doesn't change assumptions or provide radical new ways of looking at and doing things. I'm hearing different versions of what a fourth 'wave' might be like, but I can't see that any of

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the things that are being proposed as significantly changing the way systems thinking has been developing. It might be a bit imperialistic, but I think we can understand these versions of a fourth 'wave' in the context of critical systems thinking—which is this broad attempt to give respect to the wide variety of systems approaches that exist, whatever traditions of thought they originated from.

Matt Lloyd and Rajneesh Chowdhury

Sure. You've mentioned about Indigenous thinking and alternative ways of knowing. A lot of people are talking about consciousness. In my (Rajneesh Chowdhury) papers, I talk about consciousness from the perspective of Hindu philosophy. These alternative forms of knowledge surely have a lot to add to systems thinking. What do you think about these new contributions?

Mike C. Jackson

Shall I answer that by just talking about non-Western systems thought?

Matt Lloyd and Rajneesh Chowdhury

Yes, alternatives to the rational-analytical approach?

Mike C. Jackson

Yes, I do think that Eastern traditions of systems thinking—Chinese, Indian and so forth—and Indigenous ways of thinking have much to contribute to the systems approach and that these contributions were colonised and have been marginalised for too long. To take an example, I learned a lot from Tyson Yunkaporta's (2019) book 'Sandtalk' which expresses the systemic thought of First Nation people in Australia. I've tried to respect that, and insights from Canadian First People, in my new book (Jackson 2024). But it's a mistake to think that Western thought has always been rational-analytical. The guest speaker at the Mike Jackson Lecture in 2024, Charles Foster, traces, in one of his books (Foster 2021), the development of human consciousness over 40 000 years. He shows that Upper Palaeolithic thinking was much more systemic in nature. And people in the West did think more systemically until the scientific revolution led to the triumph of reductionism. Everything then went in one direction, with Galileo's and Descartes' and Newton's mechanistic thinking pushing holism to the sidelines. It survived, of course, in the work of the romantic poets, in Goethe's science and in Hegel's philosophy. But reductionism came to dominate the way people think and inevitably, therefore, the way they treat the natural world, other species and, sometimes, other human beings.

And so yes, there are clearly important systems lessons to be learned from other traditions and times. It is a matter of redressing the balance. Yunkaporta and Foster do not want to give up all the benefits that have been gained from the scientific and technological revolutions. Old ways of thinking and acting cannot provide ready solutions to the problems that the world faces today, having moved on historically, economically, politically, culturally and in terms of the state of the environment. We need systems thinking that is forged for and relevant to the current era. Which should, of course, draw upon the strengths of all the different traditions.

Matt Lloyd and Rajneesh Chowdhury

So why is it such a challenge to get systems thinking taught in educational institutions and adopted in policy arenas? And what needs to be done to popularise systems thinking?

Mike C. Jackson

I think it's a big problem that it's not well established in universities. You would expect me to say that being a systems thinker and having spent much of my career as an academic and as Dean of a Business School. Systems thinking is not well established in universities and certainly not at prestigious universities. Universities organise themselves around disciplines. Systems thinking is a transdisciplinary approach. Academics advance in their careers by specialising in particular disciplines, so this works against systems thinkers. Universities remain, for the most part, ivory towers. Academics aren't required to go out and learn from the real world. Students are educated in a similar kind of way, in theory rather than in practice. They don't usually go out into the world to learn their lessons. The need for a transdisciplinary approach to tackle real-world problems bypasses academics and students.

Now, contrast that to the way Dewey, the pragmatist philosopher and educator, said people should be taught. For him, it was all about transdisciplinary education out in the world. He said, you know, let's have cooking as part of the syllabus and teach people about the physics of cooking, the chemistry of cooking, the biology of the plants used, the processes and the need for cooperation. Everything comes together in cooking. You need a transdisciplinary approach in dealing with issues in the world. Universities thrive on disciplines and are allowed to get away with it. Systems thinking not being established in prestigious universities means it is ignored by power brokers. It is a massive problem that needs to be overcome.

Ironically, people talk about complexity as the main problem we face nowadays. That's the case whether you listen to the United Nations, UNESCO, the World Health Organization, businesses, central and local government or charities. Complexity is the issue. And they all also say that systems thinking is the best approach to tackle complexity because it can deal with interconnectivity and different viewpoints better. The demand for systems thinking has never been so great because everybody is confronted by increasing complexity and all recognise that traditional ways of thinking and doing things aren't working so well. But, for the reasons I've mentioned, we're not producing a ready supply of people trained in systems thinking. There is a big capabilities gap—a huge demand for capable systems thinkers and a lack of supply. Things are slowly beginning to happen. In the United Kingdom, for example, the Government has released

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four documents trying to ensure that civil servants are equipped with systems thinking approaches. The Systems Thinking Apprenticeship has been approved by the UK Government and more universities are coming on board to teach the apprenticeship. A Systems Thinking Working Group has been established by the Government Office for Science and SCIO ('Systems and Complexity in Organisation'), the professional body for systems thinkers in the United Kingdom. It's looking at ways of introducing systems thinking more into policymaking in the United Kingdom. So that's just a few examples, in one country, where things are beginning to happen.

Matt Lloyd and Rajneesh Chowdhury

So, you are optimistic about the future of systems thinking in the next decade. How do you see things changing in the world of systems thinking in the next decade?

Mike C. Jackson

Yeah, I am. I am optimistic for the reasons that I've just set out. People are increasingly seeing complexity as the issue. They're beginning to see that you cannot address complexity through individual disciplines, you need a transdisciplinary approach which recognises and employs the kinds of methodologies and concepts that systems thinking brings to the table. And what is the alternative? Things are not going well at the moment, are they? If we don't manage to ensure that systems thinking has equal prominence with classical science and traditional ways of thinking about and doing things then I can't see us addressing appropriately the dangerous issues that we confront in the world, and improving things in the way that we all hope to do. So out of necessity, I think people will see the need for systems thinking. I am optimistic. People have the capacity to see why things are going wrong and that we can do better. And once they are convinced of that, people will begin to demand that systems thinking is researched, taught in universities; more systems programmes will be established. The lead will need to come from government and business, from all those who need systems thinking to cope better with the complexity they face. And so, you know, there is the potential here for an explosion which will bring about a different way of thinking and doing. Things can be transformed.

Matt Lloyd and Rajneesh Chowdhury

This also relates to the previous point on how we increase the adoption of systems thinking in various institutions. We think that language plays a very important role in this. The reason we stress this is because the language that systems thinking scholars use, often, seems inaccessible—recursions, autopoiesis, structural coupling and so forth. Such language is not used in general management and administration. Russ Ackoff said that until we communicate with our potential users in a language they can understand, they and we will not know what we are talking about. So, what do you think of this comment and how do you think we can make systems thinking more accessible in terms of language? Or should we?

Mike C. Jackson

Well, there are some convoluted systems traditions, like Maturana and Varela's second-order cybernetics, where some of those concepts you mention come from. But, in general, that's not my favourite Ackoff quote. I mean what it says is clearly right. There must be proper communication between those who are developing the new systems ideas and those who can benefit from using them. As I've said before, critical systems thinking doesn't want to replace decision-makers. It doesn't see itself as a science with knowledge of the truth. It sees itself as something which can build upon the experience of people who know a situation they're engaged with and it believes it can help them think more roundly, more broadly, about that situation. It can perhaps help them think more deeply, challenge the way they're thinking about it currently and offer decision-makers alternative ways of thinking about and acting in the situation they find themselves in.

So, the intent of the Ackoff quote is correct. On the other hand, it seems clear that our current language is failing us. The concepts we traditionally rely on are unable to get to grips with the kinds of problems that decision-makers are facing. In those circumstances, clearly, you must introduce alternative concepts which open minds to new ways of thinking. If this is difficult, then so be it. There's no point in it being easy if it just leads people to think the same way. We have to introduce new concepts and trust that decision-makers discover that they have resonance for them in the particular situation in which they find themselves and allow them to gain better traction in changing things in the way they want to change things. I'm not going to be apologetic about introducing new concepts, some initially difficult, if they encourage new ideas to flourish, different worldviews to be entertained, and some new ways of thinking about the problems we face, together with new ways of acting to improve the world in which we live.

Matt Lloyd and Rajneesh Chowdhury

Wonderful. Our last question concerns what your advice would be to students, or people in general, who are just starting their journey with systems thinking.

Mike C. Jackson

OK, my first point is that I still believe, after all these years, that it is the best approach for linking theory and practice. That's where its power comes from. There are lots of different disciplines—sociology, philosophy, organisation theory and so forth—that are full of grand theories but don't know how to translate them into ways which can benefit human beings and the planet we live on. On the other hand, you have people with an understanding of the situation which they find themselves in but are locked into certain ways of thinking that prevent them improving things in the way that they might want to improve them. Systems thinking helps you, through its methodologies, to translate theories into practice and to learn from interventions at the practical level in a way that enhances the theory that you're working with. Theory and practice can be improved together

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over time. Geoffrey Vickers, who wrote some important books on systems thinking, which heavily influenced Peter Checkland (incidentally Vickers himself was influenced by Boulding), said that he hadn't come across an approach and language as good as systems thinking for helping him to understand the things that he was concerned about in his jobs in public administration and in business. He held significant roles in different industries. He'd never found anything like systems thinking that enabled him to understand the issues that he was facing. It gave him a new way of thinking about things. It's this link between theory and practice, through methodology, that systems thinking provides and which for me is an important thing about it that makes it so powerful and worth spending a career doing.

Matt Lloyd and Rajneesh Chowdhury

Just to get a sequence right—Boulding influenced Vickers who influenced Checkland? And then, we just talked about the need to spend time learning about systems thinking.

Mike C. Jackson

Yeah, the sequence is right. Regarding understanding the true value of systems thinking then, I would say, that you must do the work. There's an apprenticeship to be served. A lot of people just talk the talk. Systems thinking is very popular now, for example, in consultancy. People want to present themselves as experts and give courses. To me, this sells the systems thinking tradition short. You have to put the hours in, as with anything. I think somebody's written a book about how many hours you need to put in to become an expert at anything.

You have to do the work; otherwise, you just end up reinventing the wheel. It can be quite annoying for someone like me to see some so-called 'new' systems principle being reinvented and discussed when it was clearly discredited, say in the tradition of cybernetics, some 50 years before. Systems thinking has a history, it has developed a variety of well-tested methodologies, many useful concepts. It only reveals its full power once you have put in the apprenticeship. It's worth doing that. For people starting out and wanting to become experts in systems thinking, there are all kinds of careers to be made. There are a lot of subdisciplines in management—supply chain management, logistics, project management, evaluation, information systems, knowledge management and AI—that could constitute useful practices but currently lack the necessary theory. I think systems thinking can supply a theory for these practices. The way to enhance them is to embed them in a systems thinking approach. And not just management, of course, other obvious ones like sustainability research, ecology and so forth.

Systems thinking can provide an overall framework whereby you can carry through the translation of ideas into practice and, learning from the practice, enhance the theory. Many of the subdisciplines remain in thrall to mechanism; they need liberating. There are real careers to be made for young academics who want to take their own disciplines forward, especially in the case of applied disciplines, by using system thinking approaches.

5 | Conclusion

Matt Lloyd and Rajneesh Chowdhury

What about the future for yourself?

Mike C. Jackson

Well, systems thinking has provided a lifelong journey for me. It's exciting when you're young, you learn about it, and it gives you a whole new perspective on the world. During your career, if you're a manager or decision maker or whatever, it helps you to do things in different and I think better ways and to be more reflective. And when you get old, it helps you to think of your place in the universe. Eastern religion is often very systemic for associated reasons. My favourite philosopher at the moment is Spinoza. Spinoza is a systems thinker par excellence. He's the philosopher who, many other philosophers say, got as close as you can possibly get to being right. And Spinoza is a systems thinker. So, when you get old and you're contemplating your life and your place in the universe, and how everything fits together, systems thinking is still helpful to you. You can make a life of systems thinking.

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References

Beer, S. 1981. *Brain of the Firm*. 2nd ed. Chichester, UK: Wiley.

Boulding, K. E. 1961. *The Image*. Ann Arbor, MI: University of Michigan Press.

Burrell, G., and G. Morgan. 1979. *Sociological Paradigms and Organisational Analysis*. London, UK: Heinemann.

Checkland, P. B. 1981. *Systems Thinking, Systems Practice*. Chichester, UK: Wiley.

Churchman, C. W. 1979. *The Systems Approach*. 2nd ed. New York: Dell Publishing.

Flood, R. L., and M. C. Jackson. 1991a. *Creative Problem Solving: Total Systems Intervention*. Chichester, UK: Wiley.

Flood, R. L., and M. C. Jackson, eds. 1991b. *Critical Systems Thinking: Directed Readings*. Chichester, UK: Wiley.

Foster, C. 2021. *Being a Human*. London, UK: Profile Books Ltd.

Jackson, M. C. 1982. "The Nature of Soft Systems Thinking: The Work of Churchman, Ackoff and Checkland." *Journal of Applied Systems Analysis* 9: 17–28.

Jackson, M. C. 2019. *Critical Systems Thinking and the Management of Complexity*. Chichester, UK: Wiley.

Jackson, M. C. 2020. "How We Understand 'Complexity' Makes a Difference: Lessons From Critical Systems Thinking and the COVID-19 Pandemic in the UK." *System* 8, no. 4: 52. https://doi.org/10.3390/systems8040052.

Jackson, M. C. 2023. "In Search of a Golden Mean for Systemic Evaluation: A Response to Michael Quinn Patton." *Systems Research and Behavioral Science* 40: 636–638.

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Jackson, M. C. 2024. *Critical Systems Thinking: A Practitioner's Guide*. Chichester, UK: Wiley.

Midgley, G. 2000. *Systemic Intervention: Philosophy, Methodology, and Practice*. New York: Kluwer/Plenum.

Mingers, J. 2014. *Systems Thinking, Critical Realism and Philosophy*. Kindle ed. London, UK: Routledge.

Morgan, G. 1986. *Images of Organization*. Beverly Hills, CA: Sage.

Patton, M. Q. 2023. "Letter to the Editor." *Systems Research and Behavioral Science* 40: 633–635.

Pepper, S. C. 1942. *World Hypotheses: A Study in Evidence*. Berkeley, CA: University of California.

Ulrich, W. 1983. *Critical Heuristics of Social Planning*. Bern, Switzerland: Haupt.

von Bertalanffy, L. 1971. *General System Theory*. Harmondsworth, UK: Penguin.

Yunkaporta, T. 2019. *Sandtalk*. Canberra, Australia: Text Publishing.

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